SafeAd AI: A Multimodal Trust and Safety Framework for Early Detection of Harmful Advertisements and Age-Aware Content Moderation on Social Media Platforms

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Abstract

The rapid growth of social media advertising has increased the challenge of detecting harmful, misleading, and age-inappropriate ads before they reach users. Because existing systems rely on isolated image or text analysis, they fail against multimodal evasion tactics and user age misrepresentation. To address this, we propose **SafeAd AI**: a unified trust and safety framework integrating Computer Vision, NLP, OCR, and explainable AI to evaluate visual elements, text, metadata, and age-confidence indicators pre-publication. By generating transparent risk scores and detecting age misrepresentation, the system protects vulnerable minors, optimizes human moderation workflows, and ensures a safer digital advertising ecosystem.

Problem Statement

Development of a multi-modal AI-based trust and safety framework for social media platforms that detects harmful advertisements and performs age aware content moderation by integrating visual, textual, and contextual information to prevent inappropriate content from reaching users.

Objectives

- **Develop a Multimodal Pipeline:** Synthesize image features, textual semantic contexts, and embedded graphic text via OCR into a central assessment matrix.
- **Automate Proactive Ad Filtering:** Catch scams, explicit themes, and deceptive marketing materials *pre-publication*.
- **Implement Age-Confidence Analytics:** Estimate potential age misrepresentations to restrict adult-targeted marketing campaigns from reaching minors.
- **Establish Transparent Risk Scores:** Build an Explainable AI (XAI) output module highlighting precise policy violations for transparency.
- **Incorporate Human-in-the-Loop (HITL):** Efficiently flag and forward high-uncertainty or borderline violations to human teams to optimize manual review queues.

Alignment with UN Sustainable Development Goals (SDGs)

SafeAd AI maps directly to the following United Nations Sustainable Development Goals:

SDG 3: Good Health & Well-Being

- Protects the mental health of children and teenagers by preventing exposure to explicit, harmful, or predatory content.

SDG 9: Industry, Innovation & Infrastructure

- Drives technological innovation by introducing state-of-the-art multimodal explainable AI architectures to safety platforms.

SDG 16: Peace, Justice & Strong Institutions

- Fights online fraud, deepfake scams, and hate speech, fostering a more transparent, safe, and just digital community.

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Thank You!

Questions & Discussion